SHRIKANT SHARMA

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Data Scientist | Applied AI, LLM Evaluation & Machine Learning | Python, SQL, LangGraph, AWS

Data Scientist with ~8 years across regulated pharma and financial services, combining production machine learning with recent depth in agentic AI and LLM evaluation. Ships models end to end: benchmarked selection, calibration, deployment as monitored services, and drift detection. Builds the evaluation frameworks that prove systems actually improve (LLM-as-judge, faithfulness, regression suites), and applies causal inference to separate real effects from confounding, across 3 shipped systems.

SKILLS

Machine Learning & AI: Machine Learning, LLM Evaluation (LLM-as-a-Judge, Faithfulness, precision@k, MRR), Agentic AI, RAG & Vector Search, Causal Inference (PSM, G-Computation, IPW), Experiment Design & A/B Testing, MLOps & Drift Monitoring, Feature Engineering, Guardrails, Prompt Engineering, NLP, Gradient Boosting, Deep Learning, Anomaly Detection, Segmentation, Time-to-Event Forecasting, Calibration, Explainability (SHAP)

Languages & Libraries: Python (Pandas, NumPy, Scikit-learn, XGBoost, PyTorch, Hugging Face Sentence Transformers, FAISS, SHAP, RAGAS, LangGraph, DoWhy, lifelines, PySpark, Matplotlib), SQL, R

Cloud & Data Platforms: AWS (S3, EC2, Lambda, ECR), Databricks, Snowflake

Tools & Platforms: MLflow, Docker, Git, FastAPI, Apache Spark, Claude Code, Model Context Protocol (MCP), Streamlit, Tableau, Power BI

EXPERIENCE

Amgen | Data Scientist **Sept 2022 – May 2026**

- Led an initiative to apply NLP to compliance document processing, developing and deploying a text classification model (TF-IDF, Random Forest; Python, Scikit-learn), engineering the feature pipeline and evaluating classifier performance on held-out data before rollout; cut manual document categorization by 25%.
- Contributed to enterprise AI direction at Amgen through 2 internal GenAI hackathons, prototyping generative AI applications for compliance and audit workflows.
- Designed a multivariate statistical risk model in Python (Pandas, NumPy, Scikit-learn) scoring 65+ countries across 20+ indicators, with feature normalization, weighting, and tiered classification. Rebalanced weights from 50/50 to 75/25 toward quantitative signals and expanded output from 3 tiers to 5, surfacing 3 high-risk markets that had gone undetected. Detected degradation across annual cycles by comparing score distributions, then rebuilt and revalidated. Presented methodology to Directors and Executive Directors, informing resource allocation across 3 regions.
- Implemented statistical anomaly detection for transactional monitoring across 5 payment processes on 50K+ records (US, ELMAC, JAPAC), with ongoing performance monitoring; reduced manual review volume by 35%.
- Architected production data pipelines on AWS and Databricks behind the Amgen CMS Open Payments federal submission (~509K records, ~$630M), with automated SQL validation gates holding the failure rate under 0.5% across 4 annual submission cycles. Created year-over-year Tableau dashboards and automation that cut cycle time from 2 weeks to 4 days.
- Delivered 3 concurrent projects on schedule while owning end-to-end knowledge transfer of analytics workflows during organizational restructuring; mentored 3 junior data scientists on Python automation, modeling practice, and dashboard development.

Nugget.ai | Data Scientist Intern **Jan 2022 – May 2022**

- Trained models in Python (Scikit-learn) to predict candidate-role fit from behavioral assessment data, engineering 20+ features spanning structural, linguistic, behavioral, and content-based signals. Compared Logistic Regression, Random Forest, and Gradient Boosting with cross-validation, improving matching accuracy by 15% over the existing platform baseline.

Atos Syntel | Data Scientist | *Pune, India | Client: American Express* **Feb 2018 – Dec 2019**

- Engineered production risk scoring pipelines in Python and SQL over 10M+ transactions, engineering behavioral features around spending velocity, geographic patterns, and customer segmentation that fed downstream fraud and credit risk models and reviews.
- Designed Tableau dashboards consolidating 5+ source systems into a single executive view used in the weekly credit risk review, supporting mitigation decisions totaling over $2M and cutting reporting time by 30%.
- Owned the A/B testing framework in Python, SQL, and Git for credit risk strategy experiments, handling experiment design, power-based sample sizing, and significance testing with multiple-testing correction across 12+ cycles per year.

Atos Syntel | Data Analyst | *Chennai, India* | *Client: American Express* **Mar 2016 – Jan 2018**

- Developed the SQL data pipelines processing 5M+ financial transaction records monthly, with automated validation logic and data quality checks that held accuracy at 99.5% for regulatory reporting.
- Automated 15+ recurring reports using SQL and Python, freeing up 30+ hours of manual work each month so the team could focus on analysis instead of data prep.

APPLIED AI & MACHINE LEARNING PROJECTS

Clinical Trial Intelligence System | *Python, LangGraph, Sentence Transformers, PubMedBERT, Cross-Encoder, Groq Llama 3.3 70B, FAISS, RAGAS, Streamlit, Git* **May 2026**

github.com/Shrikant-Sharma/clinical-trial-rag | shrikant-clinical-rag.streamlit.app

- Designed and deployed an agentic Corrective RAG system on LangGraph over 484 ClinicalTrials.gov oncology protocols (3,264 chunks): multi-step workflow with LLM-as-judge grading, query rewriting, conditional routing, and bounded retries. 3 refusal gates correctly reject adversarial queries scoring 0.91 cosine similarity that a single threshold missed.
- Created the evaluation harness that measures it: a curated 25-query in-domain and adversarial set, RAGAS faithfulness scoring, retrieval metrics (precision@k, MRR), and an LLM-judged regression suite rerun after every change; the harness quantified a 34-point refusal-rate reduction from a single architectural change.
- Engineered the retrieval stack from measured failure: compared 3 chunking strategies, switched to domain-tuned PubMedBERT embeddings after a general-purpose model scored relevant oncology documents at 0.24 similarity, and added a cross-encoder reranker for 2-stage retrieval, halving in-corpus refusal rate (67% to 33%). Deployed live on Streamlit with graceful rate-limit degradation.

Patient Risk Stratification Pipeline | *Python (Scikit-learn, XGBoost, PyTorch, SHAP, DoWhy, lifelines, scikit-survival), MLflow, FastAPI, Docker, AWS (Lambda, ECR, Mangum), Git* **May 2026**

github.com/Shrikant-Sharma/patient-risk-stratification | Live API on AWS Lambda

- Delivered an end-to-end ML pipeline for clinical heart disease risk stratification on UCI Heart Disease (303 patients, 13 clinical features); selected XGBoost from 5 model families via MLflow experiment tracking and 5-fold cross-validation, reaching 0.95 AUC-ROC and 0.93 recall (precision 0.81) at the clinical operating threshold, identifying 93 of 100 high-risk patients for early review.
- Added SHAP TreeExplainer for global and per-patient feature attributions critical for clinical decision support, validated probability calibration at Brier 0.092 with reliability curves, and tuned the decision threshold to prioritize recall while maintaining precision >= 0.80.
- Extended the pipeline to causal inference and survival analysis on the NHEFS cohort (1,629 subjects); estimated treatment effects via propensity score matching and G-computation with DoWhy (3-way convergence within 0.2 kg of the IPW reference, 3 sensitivity tests passing) and modeled 10-year mortality with Cox PH and Random Survival Forest at 0.80 concordance, revealing that unadjusted log-rank significance (p=0.005) was pure age confounding.
- Deployed live as a public REST API on AWS Lambda via Amazon ECR and Mangum (ASGI-to-Lambda adapter); multi-stage Docker build, EventBridge cold-start warmer holding warm latency at ~275ms, FastAPI Swagger UI for interactive testing, and monitoring of input and prediction distributions for drift.

Payment Transaction Anomaly Detection | *Python (Pandas, Scikit-learn, PySpark), Snowflake, Tableau, Git* **May 2026**

github.com/Shrikant-Sharma/pharma-compliance-spend-analytics | public.tableau.com/app/profile/shrikant.sharma

- Developed a payment-transaction anomaly detection system on 16M+ real payment records ($13B) for large-scale transaction-transparency review, engineering 5 entity-level features across 989K sampled transactions and 289K counterparties.
- Applied within-segment z-scores plus global IQR with a monetary floor and concentration logic, flagging the top 1.67% of entities as a HIGH-risk triage queue for investigator review.
- Layered Isolation Forest anomaly detection on top; reconciliation against the rule-based flags surfaced 3 distinct anomaly archetypes (caught by BOTH methods, rules-only, and model-only), revealing suspicious patterns no single method catches alone.
- Engineered the same feature pipeline across 3 execution backends (Pandas, Snowflake, PySpark) with row-level equivalence verified; Snowflake SQL surfaced that the top 4 counterparties concentrate 59.7% of segment payments, visualized in 2 interactive Tableau dashboards.

EDUCATION

Clark University, Worcester, MA | **Master of Science in Data Analytics**, GPA: 3.6/4.0 **May 2022**

Sant Gadge Baba Amravati University, India | **B.E. in Electronics & Telecommunication**, CGPA: 7.4/10.0 **Feb 2016**

ACHIEVEMENTS

Acclaim Award (3-Star) | Amgen | For data science delivery and cross-functional collaboration **Feb 2025**